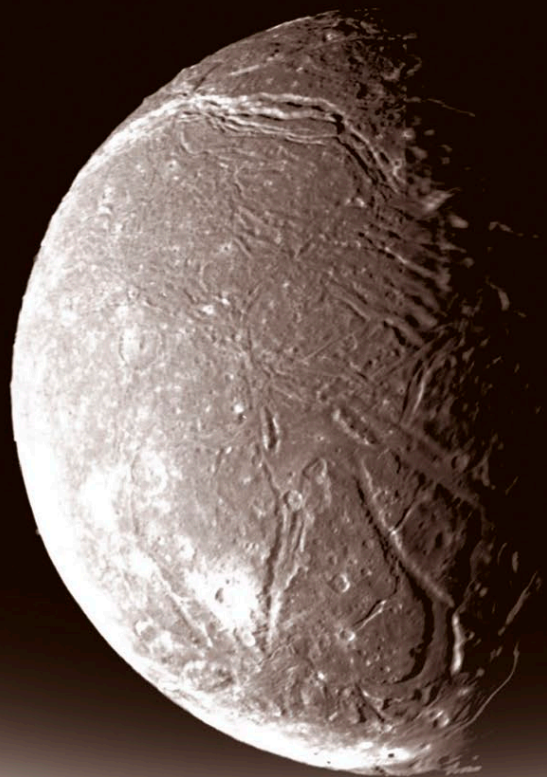




MAJOR MOON

Ariel

DISTANCE FROM URANUS 118,620 miles (191,020 km)

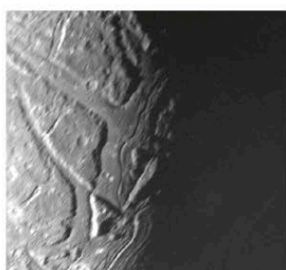
ORBITAL PERIOD 2.52 Earth days

DIAMETER 722 miles (1,162 km)

Ariel and Umbriel (below) were both discovered on October 24, 1851, by the English brewer and astronomer William Lassell (see p.207). Ariel is named after a spirit in Shakespeare's play *The Tempest*. Of the four largest moons, this is the brightest, with the youngest surface. It has impact craters,

COMPLEX TERRAIN

The long, broad valley faults in Ariel's southern hemisphere are filled with deposits and are more sparsely cratered than the surrounding terrain.

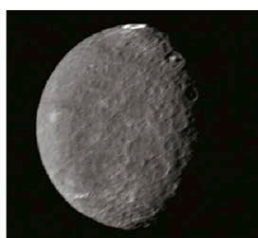


VOYAGER 2 MOSAIC

Four Voyager 2 images were combined to produce this view of Ariel. Kachina Chasmata slices across the top and the Domovoy Crater is on the left, below the center. Below and to its right is the 30-mile (50-km) wide Melusine Crater, which is surrounded by bright ejecta.

but these are relatively small—many are just 3–6 miles (5–10 km) wide. Domovoy, at 44 miles (71 km) across, is one of the largest. The sites of any older, larger craters that Ariel once had have been resurfaced. Long faults that formed when Ariel's crust expanded cut across the moon to a depth of 6 miles (10 km). One fault, Kachina Chasmata, is 386 miles (622 km) long. The floors of such valleys are covered in icy deposits that seeped to the surface from below.

observing such a small, distant moon that reflects little light. Voyager 2 revealed a world covered in craters, many of which are tens of miles across. Unlike Ariel, Umbriel appears to have no bright, young ray craters, which means that its surface is older. There is no indication that it has been changed by internal activity. Umbriel's one bright feature, Wunda, is classified as a crater, although its nature is unknown.



SOUTHERN HEMISPHERE

Umbriel is almost uniformly covered by impact craters. Its one bright feature, the 81-mile (131-km) wide Wunda at the top of this image, is unfortunately virtually hidden from view.

MAJOR MOON

Titania

DISTANCE FROM URANUS 270,700 miles (435,910 km)

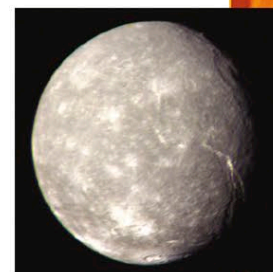
ORBITAL PERIOD 8.7 Earth days

DIAMETER 979 miles (1,578 km)

At a little less than half the size of the Moon, Titania is Uranus's largest moon. This rocky world has a gray, icy surface that is covered by impact craters. Icy material ejected when the craters formed reflects the light and stands out on Titania's surface. Large cracks are also visible and are an indication of an active interior. Some of these cut across the craters and appear to be the moon's most recent geological features. They were

FULL DISK

At top right is Titania's largest crater, Gertrude, which is 202 miles (326 km) across. Below it, the Messina Chasmata cuts across the moon.



probably caused by the expansion of water freezing under the crust. There are also smooth regions with few craters that may have been formed by the extrusion of ice and rock. Titania was discovered by the German-born astronomer William Herschel on January 11, 1787, using his homemade 20-ft (6-m) telescope in his backyard in England.

MYTHS AND STORIES

QUEEN OF THE FAIRIES



Titania and Oberon are the king and queen of the fairies in William Shakespeare's play *A Midsummer Night's Dream*. After a disagreement, Oberon squeezes flower juice into Titania's eyes as she sleeps so that on awakening she will fall in love with the next person she sees. Titania wakes and falls in love with Bottom, the weaver (seen here in a movie still from 1999), who has been given an ass's head by the impish sprite Puck.

MAJOR MOON

Oberon

DISTANCE FROM URANUS 362,370 miles (583,520 km)

ORBITAL PERIOD 13.46 Earth days

DIAMETER 946 miles (1,523 km)

Oberon was the first Uranian moon to be discovered—William Herschel observed it before spotting Titania. It has an icy surface pockmarked by ancient impact craters. There are several large craters surrounded by bright ejecta rays. Hamlet, which is just below center in the Voyager 2 image below, has a diameter of 184 miles (296 km). Its floor is partially covered by dark material, and it has a bright central peak. A 4-mile (6-km) high mountain protrudes from the lower left limb of the moon.



ICY SURFACE

OUTER MOON

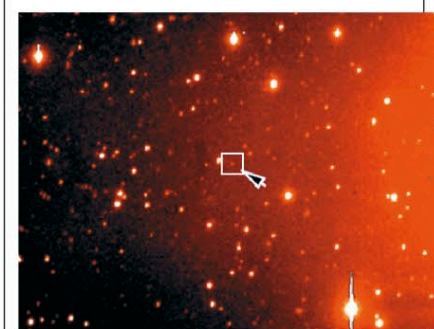
Caliban

DISTANCE FROM URANUS 4.5 million miles (7.2 million km)

ORBITAL PERIOD 579.5 Earth days

DIAMETER 60 miles (96 km)

Caliban and another small moon, Sycorax, were discovered in September 1997. Both moons follow retrograde and highly inclined orbits. Sycorax is the more distant of the two, at 7.6 million miles (12.2 million km) from Uranus. They were the first of Uranus's irregular moons to be discovered and are believed to be icy asteroids that were captured soon after the planet's formation.



CALIBAN DISCOVERED

Caliban lies within the square outline in this image, which was taken using the Hale telescope at Mount Palomar, California. The glow on the right is from Uranus, and the bright dots are background stars.